ActInf Textbook Group ~ Cohort 5 ~ Session 21 (Chapter 9, Part 2) ~ 3/25/2

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 21 (Chapter 9, Part 2) ~ 3/25/2024 | 55:37

all
right welcome back cohort
5 we are in
our second discussion on chapter nine so
Andrew or anyone else who wants to write
a question or raise their hand let's
just jump into
it
okay I just heard the recording in
progress thing maybe there's something
about Pete's Coffee that causes a time
dilation but either way now we're in
chapter nine so where shall We
Begin yeah I was going to say if anyone
has any particular points they want to
get into with this
chapter happy to jump into those
otherwise um if anyone's like you know
only able to make this meeting ween able
to make the last one I'm happy to kind
of give a general overview of what's
going on
here uh if you don't mind a general
overview would be good to start
with if other s
c yeah go for it previous meeting is at
11:00 in the at night in my time so I I
haven't been uh able to make them so
General overview would be really helpful
for me if if it's a
possibility yeah great um yeah let's do
that and there is a lot going on in this
you know this chapter this is sort of
the aside from chapter 6 which gives us
a general recipe for

composing uh gener generative models and
kind of the
more General method you might take here
in actually applying active inference in
the second half of the book um recall in
chapter 7even we're given the um PDP
model how to construct that chapter
eight we're given more continuous models
and then here we bring it back to okay
so how do we actually use these models
and construct them in a way that we can
apply them to empirical experiments
where we're actually collecting real
data um and so we're we're moving beyond
the range of just purely building models
and running simulations to collecting
empirical data such as like
neurophysiological data could be EEG it
could be just
observing um behavior of anywhere from
human beings to to rats or if you're in
more of a an engineering setting you
might be looking at other kinds of data
that's being
extracted and you just basically we want
to figure out how is that what are the
what's the
process uh that is generating that data
which of course we may never know um but
we can attempt to model it so um yeah
this
chapter it's it's quite dense I'll be
honest whenever we incorporate
everything um but it does give us a nice
full flow of how to apply these methods
to empirical data um so first of all in
the introduction there's hints at what
we'll find in the rest of the chapter um

we're introduced to metab basian
analysis which is where we actually view
this as the agents subjective model
versus our own objective model that
we're building of the agent subjective
model that kind of uh carries into the
active inference ontology right just we
do have to account for the fact that we
are sort of the quote unquote objective
scientists in this situation we're
attempting to model the behavior um and
other parameters of like an a separate
being from ourselves like in a teamas
we're trying to you know understand the
subjective model of a rat which we don't
have you know collecting data and
applying analysis to see if we can do
something like approximate it so here
our goal is to what we would say is
recover or infer the unknown parameters
of they phrase it as the subject's brain
its subjective model using our own
objective just by drawing on the
behavior in our observed data and then
we uh invert our objective model to
infer those parameters of the subjective
model and then finally by doing so we
can test and compare hypotheses for
example we might find multiple models
that we could build we could compare
them and um and see which one holds best
to like the behavioral outcomes that
we're seeing in the data and then they
they have another concept called
computational
phenotyping um this is where we kind of
by by recovering model parameters prior
beliefs of our agent we could actually

like kind of
classify um our our the individuals
we're studying whether it be multiple
rats and a teamas it could be clinical
patients and more of a you know um uh in
the study of Psychiatry but we can kind
of phenotype those individuals based on
what model we end up uh developing to
try and explain their behavior um and so
yeah this chapter lends itself really
well to specifically computational
Psychiatry and neuros pychology
neurology but uh in principle all these
methods could be applied to a broader
much broader variety of phenomena
whether it
be or or even non-living
systems
um so section
9.2 um and again there's a lot going on
in this chapter so I'm going to try to
be a little bit more brief but are
introduced to um just as we've seen
basis theorem in previous chapters uh
now we have equation
9.1 which looks almost exactly
equivalent other than we've replaced uh
observations and states with uh
behavioral outcomes youu your actions to
be taken and then a Theta symbol uh
which represents the parameters of the
model and so just as we can uh update a
model's beliefs related to States
observations the relationship between
them in the form of likelihoods and
posteriors you can do we're ideally
doing that same exact thing here but now
we're mapping outcomes to mod parameters

and I'm going to pause here real quick
bres you had a
question uh yes uh actually uh I'm um
I'm just wondering so you mentioned
mention this computational yeah exactly
this one you mentioned computational
phenomenology and this subjective model
are you saying that uh I don't know
anything about it so I'm just asking uh
are you saying that somebody can
actually start uh plugging in values
inside that uh uh that Matrix that you
said like and there will be like like
what I'm saying is the values which you
encode into that and some and some
variables since you encode will be part
of that computational phenomen ology or
How does it go yeah sure um so one thing
we could do to try and answer that
question is to look at like we have in
section 9.5 just a nice like brief
series of steps for how to go about this
um and so like in Step
One quite simple we collect behavioral
data um you could even think of this as
like a a spreadsheet or something along
those lines like that kind of structure
where each row can represent an
individual like in a tm's example we
have um rat number one rat number two we
could have uh in the columns things like
what are the what are the actions or
that they took that we we saw whenever
we collected the data um what are the
associated
observations um or or or other
components of a model that we saw we
could also have like other kinds of

information like maybe the test was
being done by administering um a drug to
certain rats and not to others to try
and study the impact right so we could
have like a binary one or zero um like
if for each rat if it was administered
the drug or not um step two we formulate
um a PDP this this chapter is a little
bit on looking at discret models so a
PDP as we were introduced to in chapter
7 although later they do give us a
couple examples of um more continuous
models and what we're trying to do with
step two is that we're trying to develop
basically yeah derive a model that like
closely approximates what we see in the
data that we just collected in Step One
um but nonetheless we know that like
this is our model so this is kind of
that outer
objective model that we're seeing in uh
in the figure that Daniels pulled up
there figure 9.1 so it's kind of that
that inner box is the subjective model
we're trying to
to basically yeah devise our own
objective model of what's going on in
the rat's uh mind so to speak if we
follow te's example um so that it there
could be aspects of this that um because
your question had to do with specifying
priors um step three is specifying a
likelihood
function um and that's based on the
model that we just tried to make then
after that step four we specify prior
beliefs um and so in that step the way
they've written it we can specify prior

beliefs about the parameters in terms of
expectations and precisions often these
will be centered on zero with precisions
reflecting plausible ranges
so you might think of that is you have
your own hypothesis about what the prior
beliefs are um and you just kind of roll
with it so to speak you we're doing a
process where we're trying to build a
model that whenever we run that
model it and and run a simulation with
it hopefully it more or less
approximates the real actions uh that
say a rat and a teamas uh took
as well as like the combination between
those and the observations that the rap
received so um hopefully that provided
some kind of information or a bit of an
answer to your
question um but yeah I'm just trying to
get to yeah sorry it was a great answer
all add a little more so Pros you
mentioned
phenomenology like the kind of felt
experience and this kind of made me
think about like um three different
kinds of phenotypes that you you might
be studying you might be studying the
most grossly morphological like just the
actual bodily movement like the movement
of an arm then there's kind of cognitive
phenotypes that are internal they're not
directly observed but they're also not
like philosophical or metaphysical or
phenomenological like um the
motivational
drive so mainly I'll I think I'll just
type I I'll just type today since it's a

little strange with the internet but
suffice to say that the the
phenomenological phenotypes like the
felt experience is probably the most
difficult because now you're not just
talking about the where the arm is in
location you're not just talking about
the intent to move the arm or something
like that but you're talking about like
the experience and that is the the
challenge that many researchers are
focused on
and hey Daniel I want to give you a
heads up that uh the your audio is a
little choppy as well but I do fine I'm
just gonna type I'll just type yeah
cool super helpful
thanks
great and then yeah I'll just follow
through with um with those steps like
after you specify those prior beliefs in
step
four step five we solve for posterior
probability and model evidence so
effectively what doing is we're just
kind of continuing the process of like
how do we approximate our subjects uh
underst studies subjective model so like
here are all the different components
right all the ones that we've been
familiar with is we've proceeded through
the textbooks we have we're figuring out
likelihood functions we're figuring out
uh prior beliefs even if we're just
setting them initially as like some
value and and then need to kind of
further carve out our model from there
and maybe uh update those priors to to

better reflect what's going on in the
subjective model so um I think the word
they repeatedly use this word recover
like we're trying to recover the prior
beliefs or recover the parameters of the
model so there's no assumption that we
know what these are from the onset
instead we're we're attempting to build
a model and then kind of fine-tune
things from there to where they better
approximate what might be the actual um
priors and and the kind of likelihood
and and uh model evidence and so on that
the the agent is working with um I'll
just put one comment so so you're saying
that if um suppose we found some
empirical study and we get get the exact
value not exact value but like a like a
distribution a Goan distribution of the
prior then that will be that will be the
best that will be very helpful right
yeah presumably uh yeah if you find that
so basically we are trying to iterate
our model so that we can get to those
kind of values which have a very
accurate description of what the prior
values exactly might be yeah something
on those lines right so some some
reverse engineering yeah you're right
that's very right like we can just think
of it as like we've been introduced to
these sort of insilico synthetic models
like in for example PDP we're already
familiar at this point thanks to chapter
7even and other chapters like we'll hear
all of the components of a PDP we have
we have our D Vector which is our priors
over you know initial States um we have

our a matrix of likelihoods we have our
B Matrix of State Transitions and so on
but now when we've collected empirical
data well we only have what we as the
you know so-called objective science
scientists can observe so we can observe
uh you know an agents like a rats or
human beings actions that they've taken
and we can observe like maybe what was
going on in the generative process like
okay the mat the the rat was in a maze
where the reward was on the left right
so so we have our gnomes but they're not
uh and given and and what we observe but
they don't all match all of these
different components of the PD P so
we're attempting to kind of recover
what's missing right we're we ourselves
are under undergoing the active
inference process of trying to to infer
and learn uh what are the other
components that we cannot directly
observe of the rat's Behavior because
then that way of course it starts to
match you know anything else some other
machine learning routine or something
like we're just trying to find these
unknown parameters in what we presume to
be the rat's model subjective model
yeah thank you sure
absolutely and
then um aside from being uh either
introduced or given some kind of more
technical refresher on what what methods
we might use along the way to do this
like they dedicate an entire section to
variational Applause which kind of gives
more description on how we're using this

modified phase theorem now instead of
with States and observations we have
actions and we have um like model
parameters um aside from yeah how it it
reintroduces uh how we might yeah
recover parameters via the llas
approximation we also get a much more
description ility of applying these
methods in the first place which is in
Step six of the the overall process
uh that's introduced here and so it gets
more into you know there's a lot of
overlap between this and other more
traditional uh analytic settings where
you might be trying to do something like
a covariant analysis or or a correlation
analysis you might be trying to as I
mentioned earlier with a Tas example
where maybe a drug was administered to
certain rats and not to others let's do
some kind of group level analysis to see
what are the patterns in the
Behavior Uh and outcomes for the rats
who were administered the drug versus
those who are not right so this this is
getting into much more again like kind
of traditional settings where really
you're just applying analysis um but now
you have the the added utility that
you've Dei you've devised an entire like
in this case PDP that can
potentially like really strongly explain
a rat behavior and you have all these
different components and parameters that
you've recovered to allow for quite
broad variety of analyses Beyond just
beyond just running like a simple like T
Test or or um yeah looking at

correlation between certain behaviors
and and outcomes and that's all we have
so yeah that's that's the general
process I think that that figure 9.2
does a rather good job of trying to
visualize the the six-step process
collect data we build our own objetive
our initial objective model of what how
we think that that data is produced we
derive like a likelihood
function step four prior beliefs we set
those step five we invert the model
which we're already familiar with doing
uh previously in the book like that's
how you go from a model an agent who has
priors and likelihoods and then those
can be mapped through bases theorem to a
posterior and model evidence and so
everything actually looks kind of
structurally equivalent right um to that
equation and finally we move on to the
final analysis yeah
P um actually I have a question so so
what about if we if you are um like we
don't exactly follow the likelihood
function the way they given where
they're giving uh but we uh introduce
our own function which actually does the
likelihood mapping like not a function
or or a
methodology and uh on top of it
so so so actually the fundamentally like
we're using this Marian processes to do
the state transitions right so we have
flexibility that we use uh uh we what
I'm actually trying to say is that this
three this likelihood function right
which starts from the policy level uh

right fundamentally down to the
observation level and uh instead of
following the exact mechanism which they
give we can introduce our own methods
but
we uh do the mapping of transitions
across States and the observation
mapping like the a matrix and the B
Matrix we keep it very similar uh but
the other uh the the the higher levels
the higher uh levels
that can plausibly be changed also right
if we it's like we can apply a free
energy principle to that
also I'm just uh I'm just think on those
lines I've not been able to formalize it
very cleanly but uh I hope that in a few
weeks time I'll have more clarity um so
that's just what I was trying to
understand you know fundamentally we do
this um uh surprise minimization the the
most quote principle and then the
transition mapping but the later above
layers like say policies and affordances
we can
actually create a methodology which
fundamentally does that but not exactly
through this very uh complex basan
mechanisms uh that's just what I'm
trying to say it's like there there can
be other methods like to accomplish
those policies and
affordances
MH I'll I'll see if I can if it will
work it's a very interesting question
and and it makes me think that this is
like a policy selection but you're not
actually choosing the policy but it's

trying to develop a function that does
map from all of the model and parameters
to the choice of policy so it's kind of
like policy selection from the outside
and then we have the view from the
inside view from the outside and maybe
there are some other heris STS or rules
that could be simplifying in terms of
how we do the policy
inference
uh sorry actually I couldn't hear but
you think policy inference will be will
be from inside right or you're saying
it'll be from outside I don't understand
well Pi is usually how we think about
policy selection from the inside but
we're the behavioral experimentor we're
only
observing the the U the emitted
behaviors so we're developing a model in
step three that that is is the
distribution of the emitted Behavior
given all of the parameters many of
which the subject doesn't have access
to Daniel just a quick heads up your
your audios it's still kind of choppy
there
um but I think you were making the the
point that yeah like whenever we've
collected the data we have access
ourselves to what were the the
behavioral outcomes I.E whenever we were
looking at equation like the Tilda U
that is a series of actions um the Tilda
always Den know it's like a sequence and
then you being an individual action and
so with a policy you can think of a
policy as a series of actions like the

um know we have like a rat and a teamas
action one it takes a hint as to where
the reward is action two it goes to
where the reward is as As Told by the
hint um so so now we have like a number
of actions in a sequence like a Time
step one it did this time step two it
did this um and that can be represented
as the U Tilda which could also be
represented as an individual policy so
um a rat who really already thinks that
the reward is on the left it'll skip the
hint it'll go right to that reward right
so we see a c we can associate a certain
sequence of
actions uh
with uh you know this this actual
behavior that we see it's it's
equivalent and then whenever we're
looking that equation it's like okay
well why did the rat jump right to the
reward like what parameters are in the
model that would push it into that kind
of behavior rather than the kind of you
know safer policy of going for the hint
first um and so that's what we're trying
to kind of recover like what are those
parameters that are determining that uh
given that we only know the actions that
it took we don't know all of these
underlying parameters and and you know
cognitive parameters or otherwise we
don't necessarily know um priors we
don't necessarily know how much the rat
like another thing that we get
introduced to in this book is whenever
an agent has extremely strong priors
like a rat who just really really wants

the cheese uh and and doesn't even
really care it's it's uh not very uh
risk ofers so it doesn't really care if
it ends up you know choosing the wrong
option and getting a small shock or
something it'll run right to the cheese
um those are all yeah parameter settings
that are determining that and those are
the parameters settings that we're
trying to recover uh in the model
inversion process and that's why at step
five we get this like
reverse uh view of that same equation
instead of we have instead of having a
sequence of actions given a parameter
now we're trying to find what is that
parameter given the actual actions that
we that we
observed that kind of
tracks um I'm sorry Andrew uh
basic question could you please tell
those values what what is a p mu Tilda
that's the policy what is Theta what is
what is O and what is M yes for sure um
Theta are your model parameters it's
kind of it's sort of like a collection
of all these different parameters that
might be involved in the model right so
um a lot of notation in other areas in
active inference and elsewhere it's like
normally they would kind of put that in
bold just to to denote like oh this is a
collection of different things it's not
just like a single value right so that
might have been helpful if they used
that um but but yeah so the Theta are
just these parameters that we're trying
to recover about the model m is just a

symbol um that throughout the book
sometimes you see an M uh being included
in an equation sometimes you don't uh
but more often than not uh it's implicit
that m is almost always there m just
denotes this is a model like there's a
model in here and so it has its own
parameters it exists uh just point blank
like there's a model involved here and
so we have to say like technically all
of these parameters are conditioned on
the fact that we have
them specified maybe in a certain way
for example the M might also along the
idea that oh yeah this is specifically a
model um in the form of a PDP versus a
different kind of model um so it's it's
kind of there in part just as a reminder
and that's why it's often left out
because it's already implicit that it's
always there um and then finally the uh
Tilda o that is our sequence of
observations and so with a rat like in a
tze just as we had oh the rat like
action one the the till a youu it first
it took the hint then after that it went
for the reward we know that those are
are recorded actions that the rat took
we also know the observations that is oh
the rat whenever it took the action of
going to the hint we know that it
observed the hint um so we can track
that action number two two it went for
the reward and it it successfully got it
it got the cheese the observation in
that case at time step two is that the
the rat observed the reward so just as
we have a sequence of actions we also

have an Associated sequence of
observations and those are kind of our
two those are the two things that we
collect in empirical data by which we
now have like hopefully enough
components to try and do this model
inversion and extract those parameters
that we cannot directly observe is that
helpful uh sure can I can I just uh
clarify what I understood yes absolutely
okay sure so so what is saying is so
okay m is the model which we can think
is think of like a configuration setting
initially which we pass on to the entire
system right uh like because every model
can have a different configuration
setting maybe that that might serve what
I'm saying
um
your o o Tilda right what is it o o bar
oar is like your it's a Dela of
observation squ bar yeah and then this O
Bar is like it can be more than one
observation right it can be like a
sequence of observations but don't
exctly State how what is the length of
the what is the what is like the slot of
the sequence right so we do technically
know that like in a um like a team's
example like based on your yeah we have
a certain number of time steps so it's
like because that's that's what the PDP
is about it's about discrete time steps
so it's like time step one the rat
starts in the center like it's done
nothing there are no new observations
necessarily or maybe we say it didn't
observe anything it's just time step one

there it is time step two um it then
goes to get the H it observes the hint
so now at time step two we're able to
record here's the action it took and
here is the observation it had it
observe the hint then time step three it
decides to move to the left arm of the
maze uh and so we know that its action
it took the U is that it moved left and
then it observed a reward and so time
step 3's uh o is it observed a reward so
you can imagine that is like two vectors
almost of like here's time step one
here's time step two here's time step
three here's the vector of actions it
took from time step one two three here
are the the observations it had at time
steps one two three the threes are
typical over
here yeah in this case it's it's been
done in different ways uh for the
teammate's example for example and and
it's it's really useful because you can
do things like try and study the the
so-called explore exploit tradeoff so
imagine you you allow for four time
steps um it's where the rat starts and
then it can choose to take the hint or
knot and then it can go to the reward
well if you have four time steps if the
rat skips the hint and goes straight to
the reward well now that you have that
additional time step the rat can spend
more time with the reward and so it's
like you if you think about it in terms
of it like maximizing its utility or
reward it's like oh it gets to spend a
whole extra time step with that cheese

like that's you know great it skipped
the explore Behavior went straight to
the exploit Behavior to get as much
cheese as it could um but then you can
use that to try and study like oh this
is a very like risk-taking rat right so
it chose to do that uh versus other rats
who might be more uh conservative and
like go for the the risk uh excuse me go
for the hint first even though they're
giving up that additional time step that
they could spend getting like spending
more time with the reward and then
finally you could also have very risk
averse rats who like you know maybe they
got shocked a couple times and they have
preferences to really not get shocked uh
that that those negative consequences in
terms of its preference yeah highly
outweigh its preference for the reward
in which case you might have have a rat
who just stays in the center the entire
time or it might go to the hint as a
exploratory Behavior to gather more info
it might go to the hint and not even
trust it depending on how you set up the
situation and it might stay at the hint
and never go for the reward it just
really wants to know it's you know so is
it's as if it's
so about this stive way because it's a
real animal but like maybe it's so
scared that it just keeps attempting to
gather more info um and never goes for
the exploit Behavior so hopefully that's
maybe a kind of an illustration of like
you know we we we we can choose the
number of time steps whenever it comes

to discrete time models like this and so
we can easily just index like time step
one time step two
Etc
was that was that helpful at all per
Kush I might have been over descriptive
with the t- example there plenty of
examples actually my internet got
blocked out also I had to reconnect but
but actually I think I get get the
picture I I like this uh this that model
example yeah yeah I think it's a a
worthwhile one just because this chapter
spends time on it and that's kind of
that figure 9.2 it's like here's the
series of steps you would take if you
were doing the te's example so it's like
for anything that still seems ambiguous
or uncertain hopefully I'm giving enough
info that you would come back to the
chapter and see like okay like
I get it now with the team's example uh
Francis you had a
question yeah um so that's uh really
interesting and really helpful thank you
um am
I this maybe a kind of blindingly
obvious uh question is the focus of this
book or indeed
of the active inference program in the
general sense more
about understanding active inference as
exists uh rather than say developing
active inference as a generic
AI
technology to
solve uh
um issues where we're not

necessarily the model the generative
model is not necessarily active
related yeah I think that's a really
interesting question um and it might
[Music]
be somewhere between an answer to that
might be somewhere between uh more
objective versus versus highly opin
nated um but I'll so I'll give my answer
uh which
is as far as I'm aware and looking
at where um you know active inference
can be applied
to just kind of a ton of an immense
amount of phenomena um like here we're
given chapter n applying active
inference to
imper
where it be applied to empirical data
that we're collecting and so we're
looking at like actual agents
um active inference has many of its
foundations and you know the work of
Carl friston and others who are coming
from a
neurological
neurobiological background and so there
there is there has been kind of a big
push initially to use this for for those
fields and so computational Psychiatry
and so on and we get a variet of
examples uh of how that's been done in
this in this textbook itself in this
chapter like table 9.1 they provide a
you know an entire annotated like big
bibliography of all these different
works on uh delusions
and um

pharmacotherapy interventions and so on
but that said um know you could you
could build models that like are
entirely in silico that don't have to do
with anal I in you know animals or human
beings um like on Thursday mornings we
were using um RX and fur which is a
package developed in Julia to to try and
build in silico
experiments and so what I'm saying with
that is you could use these tools to
build AI or or otherwise different kinds
of Behavioral Systems um that you could
you know try and employ in the physical
world like you could build a you know a
robot that has its own sensory
modalities it can see you know visually
what's going on and then try and infer
what kind of actions it should take and
realiz it preference and of course like
if we build a robot unless you're trying
to actually emulate a human being or a
rat but probably not you're probably
building it to add to our reality and
can some kind of new function in society
or otherwise right in that case now it
would be more about like we're starting
with make you know simulating something
on a computer that maybe one day can be
realized in the form of like a physical
machine um so yeah it's kind of like
these two worlds right it's like we can
build active inference models to try and
better understand the world around us or
we can build models that we can then
actually employ you know in the form of
AI in in society for whatever your use
case is um yeah and and so you'll see a

lot more literature that's been
published including since the textbook
that lends itself more to the latter
approach that I described but again of
course this textbook is co-written with
friston and other folks and with a
neurobiological background so I I think
if if you really want to rely just on
this textbook you probably will get much
more of the flavor the kind of yeah
psychological
psychiatric biological flavoring of
active inference I hope that that was a
helpful uh kind of opinion that I shared
there y thanks I think that was very
helped to
clarify uh what I was feeling my way
towards thanks sure
absolutely um yeah
bries question yeah hi uh Andrew I was
uh I was since you mentioned about uh
this this Julia thing right and uh I was
planning to join but I've not been
joining because I think I will start
joining uh maybe the week after next
after I complet few things but I have a
question uh so I have never used Julia
before but I I'm seeing that it's quite
powerful and it's very fast right that's
the most uh uh it's like it's um why
people are using it but I'm also
wondering since uh does it have
Integrations with suppos say to tools
like neo4j and has anybody worked on
that like I was just seeing that they
have a driver called bold. JL which does
this I know I don't want to explore so
much it' be crazy but what I was

thinking is if if we have like
um uh okay let me let me make the
question a little bit more simpler uh
have you guys worked on any kind of
int is it entirely within the jul or do
also do some kind of Integrations
suppose
sure because like in Python the if if I
use Python for example like and I used
want to use other platforms also uh the
Integrations are like much more um like
there are like good libraries which will
help us but over here it will require a
little bit more level
of like we we've not jumped into this
knowing uh for for example Julia and
inside and
out
actually a PhD student who's working
directly with the people who uh
developed our xenfer so we do have like
a little bit of support from someone
who's been directly involved in in
building out the the package and and
resources um but the majority of us have
not even coded in Julia before so so
it's primarily like mini of us just kind
of learning this together and we're
going
through um another nice thing RX Ander
has a their website um where they've
provided a series of like different
coding examples for like different use
cases uh we did one that that that
follows um AR Roomba like a small you
know machine that cleans your home and
trying to model like an inference
process so that it can it can derive

like a likelihood of like oh if I'm on
top of tiles I must be in the bathroom
if I'm uh you know seeing a hardwood
floor I must be in the living room and
so so we just get these like nice like
uh sort of real world use cases um where
we get to follow along and figure out
how the code is functioning there and
more specifically getting back to your
question though and integration um I'm
still learning that myself so I can't
give a terribly holistic answer but what
I can say say is that anytime we've uh
run through any of these examples for
the most part we're really just using
like uh an updated Baseline you know
Julia like install that on your machine
pull it up uh we use a nice uh visual
editor called VSS code uh if you may or
may not already be familiar with that
and then um in terms of just like
bringing in the packages we're using for
the most part it's just RX and fur it's
not a ton of
well okay to be fair that has its own
dependencies as well um but yeah RX INF
fur aims to be kind of holistic like
this is the one package you have to pull
in and then maybe you pull in like an
additional like nice um like plotting
package so you can like visualize the
results of the inference process or
otherwise um that come out of ARX Ander
but yeah I think it I think ARX infer
aims to be a rather
Standalone as
asterisk um U package for for yeah
building active inference models both uh

both discreete time like pdps or even
just not even going that far and mdp
depends on your use case as well as uh
continuous models and yeah it's quite
strong and um as far as previous
approaches to active inference like
software wise we have mat lab uh which
has been more geared toward around uh
building discreete time models and uh
python where the strongest like
pre-established package for that that
being a pi
pmdp developed by uh Connor Hines and
others that's almost entirely like
exclusive to discrete time steps um so
Julia yeah something about the
combination of computational efficiency
with like oh developers who are creating
these new kind of holistic packages for
just trying
to develop everything in one go in a
single script um that's that's able to
handle continuous processes uh much
better and and and kind of more
holistically as well so um I don't know
that was a lot but I hope that
was just a followup question on Julia
like what you're doing are you like
trying to say create some functions
which will say do a thing like say
minimize surprise or say calculate varal
free energy for you guys like just
create a function which will do that are
we thinking on those lines also yeah
presumably you could Define your own um
I mean it what's really interesting to
me about Julia versus like python is
that that that you can actually like

type in a Theta and other symbols into
the code itself as to write it out and
so what I what I mean by bringing that
up is that like you could certainly um
write your own sort of objective
functions uh to
to energy
minimization um what's nice is that arcs
infer like some of its coding examples
very specifically relate to active
inference like it is explicit like an
model building uh so there are functions
like like they have this uh infer
parenthesis function where you like
here's your you plug in here's your
model that you've just built here's your
data simulate the behavior and then um
they have a single argument for free
energy true or false if you set it to
true it will minimize free energy so
it's kind of like all inhouse like uh
out of the box like if you want to do
free Min energy minimization there's um
on their website they also have
documentation for like is how they uh
like within that that that
function is the free that's being
minimized because at this point we have
a few different kinds of free energy uh
and and formulations like in this
textbook for example we're already
introduced to the difference between
variational free energy and expected
energy and since then um there there's
been more uh Research into other forms
of free energy that highly relatable uh
but nonetheless like for example trying
to combine like how do we put together

free energy into maybe an an even more
concise single
formulation with things like generalized
energy another
one be able to pronounce correctly but I
think it's birth of energy or death
energy free energy and that's that used
in this RX infer package um so
yeah suing it back up um there are ready
to go like out of the box free energy
objective functions in the package but
you could um Define them yourself as
well if you so
chose and um Daniel's making a point
with the notes one of the another
benefit of the
oener approach because I would say it's
not just a package but there's there's a
lot of documentation on how it's it's
its own approach it is um highly
relatable to the textbook we're going
through and nonetheless it it kind of
makes it attempts to make some kind of
advances on on how to do active
inference so we end up with these things
called constrained forny Factor graphs
um we already see all these different
Factor graphs in in the textbook we're
going through here um you know the the
way the PDP looks with like its
nodes edges and here's to visualize how
it works ARX and F is doing the same
thing but it's attempting to make each
each node kind of self-contained almost
such that uh you just account for its
connections with other nodes and it will
locally minimize its own free energy um
in that case like rather than looking at

a totality of the system minimizing some
broader free energy uh now you can look
at it as oh each node in this graph is
is minimizing its own free energy based
on its local connections only what's it
called this allows what is it called
what you
saying all name please what you said
what is it what is it called what's the
name yes the constrained for Factor
graph oh okay yeah and Daniel's type the
cffg yeah yeah uh for sure and then
you'll you'll see like in what's nice
another thing is that in each of the um
examples like so if you go to the RX
infer website which um you know we've
we've put links to that uh elsewhere in
the Kota but uh a simple Google search
of RX infer will will'll take you right
there as well um each of the examples
will site uh related paper so like here
here's our code example we've kept it
really concise for you but if you need
more information on the simulation we're
doing the components what the cffg looks
like for this um you can just go to the
reference paper that relates to it so I
we're part of the pull here for doing
this project is that we're also giving
feedback on the documentation for the
package since we we have this kind of
direct connection with the developers
themselves um but so we've really been
kind of scrutinizing the the
documentation a bit
check check out cffg because the the
whole idea is that you to you could you
could go far beyond the kind of uh pdps

we're seeing here uh in the textbook you
weend with a kind of a standard recipe
for building those but um a cffg because
each of the no nodes is uh minimizing
free energy locally you could end up
building in principle whatever kind of
want you could make it highly Lex all
you have to do is specify here the nodes
connects and uh you know it uses message
passing algorithms underneath to to
transfer all the messages between them
but uh yeah you could make like any kind
of arbitrary model um basically and and
like derive like free energy
minimization for each of those nodes and
then you could potentially sum those up
and look at like how the overall system
is doing and such things like that but
uh yeah the interest of time yeah I
would just recommend looking more into
the ARX and for a package and it sounds
like something that might might interest
you and then also we are posting in
theota like uh our our notes and and
what's coming up in the next
meeting the other option the active
inference as a
direct these startups using active
inference as a direct um AI uh approach
uh rather than just using active INF
rather than using active inference to
Simply understand systems which
understand yeah yeah no absolutely I
think the the two kind of approaches go
together to right because so much of
this based upon Neuroscience research
towards you can almost think of it as a
one of them is eating the other if we

can learn how
biological organisms learn for example
that that gives ideas of how to
formulate a model that itself can learn
and then moving from that to building
like a pure AI That's just using many of
those principles we've derived um for
yeah generating its own behavior despite
the fact that it itself is not a a rat
or a human being or some such thing but
yeah please absolutely feel free to to
add to the code as you like questions or
notes or thoughts think for
them so I think we we've
about sure all stop recording